Paper 32: The Zone Attractor Structure

Diffuse Noisy Attractor, Optimal Zone Width at ZONE_W≈5–7,
and FA Recovery of the $\nu = 0.003$ Failure Mode

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Abstract

Paper 31 showed that VCML zones are spatially uniform flat attractors. Paper 32 characterises the temporal attractor structure with three experiments. (i) **Attractor characterisation** (Exp A, 20 seeds, $T = 3000$–8000): the within-seed coefficient of variation of sg4 at steady state is CV = 0.44 (44%), with 56 zone-swap events across 20 seeds – a *diffuse noisy attractor*, far from a fixed point. Zone identity is maintained stochastically, not deterministically. (ii) **Extended ZONE_W sweep** (Exp C): K_{eff} vs ZONE_W is *non-monotone* with a minimum at ZONE_W $\approx$ 5–7 ($K_{\text{eff}} = 0.013$–0.014, $\Gamma \approx 8$–9). Both smaller and larger zones have higher K_{eff}, indicating an optimal zone size where vote-count efficiency peaks. The power-law fit from Paper 30 ($K_{\text{eff}} \propto \text{ZONE_W}^{-0.31}$) is invalidated: $R^2 = 0.16$. (iii) **FA recovery at** $\nu = 0.003$ (Exp B, Q21 answered): sg4 is fully restored to standard levels at FA = 2.0 ($\approx 10\times$ the standard FA). The $\nu = 0.003$ failure is energetic, not structural: insufficient consolidation force, not broken zone topology.

1 Background

Papers 28–31 have progressively characterised the zone structure in VCML: zone boundaries are stochastic Layer I objects (Paper 28); zone differentiation follows a smooth saturation law with spatial amplification Γ (Papers 29–30); zones are spatially uniform flat attractors with no within-zone gradient (Paper 31). Paper 32 now asks: *what is the temporal attractor structure?*

- Is the zone pattern a fixed point (converges and stays)?
- Is it a noisy fixed point (fluctuates around a stable mean)?
- Are there Kramers-type escape events (zones swap identities transiently)?

Paper 32 also answers the two remaining open questions from Paper 31:

- **Q20.** Does the vote-count model explain the zone-width scaling? Extended ZONE_W sweep (6 values) provides a more precise exponent and tests whether the scaling is a power law.
- **Q21.** Does stronger FA rescue the $\nu = 0.003$ low-amplitude saturation?

2 Experiment Design

Fixed parameters. All standard VCML parameters: $H = 20$, $N_{\text{zones}} = 4$, $\kappa = 0.020$, $\nu = 0.001$ (Exps A, C), WR = 2.4, HS = 2, etc.

Experiment A – Attractor characterisation (20 runs). Standard condition ($\kappa = 0.020$, $\nu = 0.001$, FA $= 0.200$, ZONE_W $= 5$). 20 seeds, $T_{\text{end}} = 8000$, checkpoints every 200 steps from $T = 3000$ (26 checkpoints). Stores: sg4_t, zmeans_t. Measures: within-seed CV of sg4 over checkpoints; zone-swap events (sg4 drops below 50% of per-seed mean); cross-seed spread of steady-state sg4.

Experiment B – FA recovery at $\nu = 0.003$ (25 runs). $\nu = 0.003$, FA $\in \{0.050, 0.200, 0.800, 2.000, 5.000\}$, $\kappa = 0.020$, ZONE_W $= 5$, 5 seeds, $T_{\text{end}} = 6000$. Note: FA $= 5.000$ caused numerical overflow (field magnitude $\to \infty$), indicating that the upper limit of safe FA at $\nu = 0.003$ lies between 2.0 and 5.0.

Experiment C – Extended ZONE_W sweep (90 runs). ZONE_W $\in \{2, 3, 5, 7, 10, 15\}$, FA $\in \{0.005, 0.020, 0.080, 0.200, 0.800\}$, $\kappa = 0.020$, $\nu = 0.001$, 3 seeds, $T_{\text{end}} = 4000$. Saturation law fitted per ZONE_W to extract $K_{\text{eff}}(\text{ZONE_W})$.

Total: 135 runs.

3 Results

3.1 Experiment A: The Zone Attractor is Diffuse

Table 1: Exp A attractor statistics. CV: within-seed coefficient of variation of sg4 at steady state. Swap: sg4 drops below 50% of per-seed mean within a 200-step interval.

Statistic	Value
Mean sg4 (T=3000–8000, 20 seeds)	89.3
Std sg4 across seeds	11.3
Mean within-seed CV	0.44
Max within-seed CV	0.57
Zone-swap events (20 seeds, 26 steps)	56
Attractor verdict	Diffuse

Table 1 shows that the zone attractor is not a fixed point. The within-seed CV of 0.44 means sg4 fluctuates by $\pm 44\%$ of its mean over the $T = 3000$–8000 window. With 56 zone-swap events across 20 seeds and 26 checkpoints (~ 520 transitions), approximately 11% of 200-step intervals see sg4 drop below half its mean – indicating frequent transient loss of zone identity.

Diffuse stochastic attractor. The correct characterisation is a *diffuse noisy attractor* (or Kramers-type metastable state):

- Zone identity persists on average (mean sg4 = 89.3 is well above zero).
- But individual realisations fluctuate enormously (CV = 0.44).
- Zones frequently experience transient collapses ("swap events") where differentiation is temporarily lost before re-establishing.

This picture is consistent with the copy-forward mechanism: zone identity is not "hard-coded" into the field – it is constantly reconstructed by the ongoing birth process. When many cells in a zone are simultaneously born from boundary neighbours, zone identity is transiently diluted. Recovery occurs as subsequent births re-establish the zone consensus.

Physical analogy. The VCML zone attractor resembles a *disordered ferromagnet* near its Curie temperature, not a ground state. The system is "above T_c" at the level of individual realisations, even though ensemble-averaged magnetisation ($\bar{\text{sg}}4$) is positive. Spontaneous symmetry breaking occurs statistically, not deterministically.

3.2 Experiment B: FA Recovery at $\nu = 0.003$

Table 2: Exp B: sg4 at $T = 6000$ vs FA for $\nu = 0.003$. 5 seeds. Standard $\nu = 0.001$ level = 89 (dashed reference).

FA	sg4 mean	sg4 std
0.050	21.0	11.3
0.200	29.5	14.1
0.800	40.1	17.1
2.000	111.6	33.6
5.000	overflow	–

Q21 answered: the failure is energetic, not structural. Table 2 shows that sg4 rises monotonically with FA at $\nu = 0.003$, reaching 111.6 at FA= 2.0 – essentially the standard level (89.3). A tenfold increase in FA over the standard value completely rescues the zone differentiation that was suppressed by excessive turnover.

This confirms the interpretation from Paper 31: the $\nu = 0.003$ low-amplitude saturation is caused by *insufficient consolidation force* relative to the birth-event disruption rate. At $\nu = 0.003$, cells are born and die 3× faster than standard, generating 3× more disruptive events per unit time. Compensating with $\sim 10\times$ stronger FA restores equilibrium.

The overflow at FA= 5.0 establishes an upper bound: the field grows as $|\mathbf{F}| \propto \text{FA}/(1 - \text{FD})$ at saturation. At FA= 5.0, $|\mathbf{F}|$ exceeds floating-point range, indicating that the stable operating range for $\nu = 0.003$ is FA $\lesssim$ 2–3.

3.3 Experiment C: Non-Monotone ZONE_W Dependence

Table 3: Exp C: K_{eff} and $\Gamma = 0.117/K_{\text{eff}}$ vs ZONE_W. $\kappa = 0.020$, $\nu = 0.001$.

ZONE_W	K_{eff}	Γ
2	0.0452	2.59
3	0.0359	3.26
5	0.0143	8.19
7	0.0133	8.78
10	0.0229	5.12
15	0.0308	3.79

Table 3 reveals that K_{eff} is *not* a monotone function of ZONE_W. It decreases from ZONE_W= 2 to ZONE_W= 7 (minimum $K_{\text{eff}} = 0.013$, maximum $\Gamma = 8.8$), then *increases* for ZONE_W= 10 and 15. The R^2 of the log-log slope is only 0.16, indicating that the power-law description ($K_{\text{eff}} \propto \text{ZONE_W}^{-0.31}$, Paper 30) was an artefact of sampling only three zone widths $(3, 5, 10)$ that happen to straddle the optimum.

Optimal zone width. The optimal zone width (minimum K_{eff}) is ZONE_W $\approx$ 5–7 at the standard parameters ($\kappa = 0.020$, $\nu = 0.001$). We can understand this as the competition between two effects:

1. **Vote-count gain** (increasing with zone width): more sites participate in building zone consensus, improving signal-to-noise. This reduces K_{eff}.
2. **Coherence-range loss** (increasing with zone width): when the zone width exceeds the copy-forward spreading range ℓ_{cf}, sites near the zone centre cannot maintain coherence with sites near the zone boundary. The zone breaks into independent patches, reducing consensus quality and increasing K_{eff}.

The optimum is where these two effects balance. At $\kappa = 0.020$, $\nu = 0.001$, the copy-forward spreading range appears to be approximately $\ell_{\text{cf}} \approx 5$–7 columns. Paper 31 showed that within-zone spatial autocorrelation is near-zero after demeaning ($\ell < 1$ column), but this measures the *gradient* coherence length, not the *identity* coherence length. Zone identity ("which zone am I?") can be maintained over longer ranges than spatial gradients.

Revision to Paper 30. The Paper 30 conclusion that $K_{\text{eff}} \propto \text{ZONE_W}^{-0.6}$ is superseded. The true relationship has a *peak* at ZONE_W$\approx$ 6 and falls off on both sides. The Paper 30 result captured the ascending flank of this relationship (narrow zones below the optimum).

4 Discussion

The zone attractor as a metastable state. The diffuse attractor (CV=0.44, frequent swap events) implies that zone identity in VCML is a *metastable* property, not a permanent one. The system has no mechanism to prevent zone collapse – it only has a mechanism to *rebuild* zone identity after it is disrupted. Mathematically, the attractor is a shallow potential well from which the system frequently escapes by Kramers-type fluctuations, then falls back in. The well depth depends on FA (deeper with stronger consolidation) and ν (shallower with faster turnover).

The attractor is not gradient-descended. The high attractor variance (CV=0.44) combined with the non-monotone ZONE_W dependence makes it clear that the VCML zone attractor is not the minimum of any smooth loss landscape. A gradient-descended fixed point would exhibit small variance (CV$\ll 1$) and would sit at the minimum of a convex function. Instead, VCML zones occupy a noisy plateau in a rugged landscape, maintained by ongoing stochastic reconstruction rather than convergence to a minimum.

The absence of gradient flow is not a deficiency – it is the mechanism. By relying on stochastic reconstruction (copy-forward births) rather than gradient convergence, VCML achieves:

- *Adaptability*: the zone pattern can change when the environment changes, because the attractor is shallow enough to escape from.
- *Mandatory turnover compatibility*: each cell that dies takes its field with it, but zone identity is distributed across all zone members and is re-learned by newborns.
- *No vanishing gradient problem*: there is no gradient, so it cannot vanish.

Optimal zone width as a system-level parameter. The optimal ZONE_W$\approx$ 5–7 is set by the balance between vote-count benefit and coherence-range cost. It is a function of κ and ν, and shifts as these parameters change (Paper 30 shows K_{eff} vs κ). In biological terms, this is analogous to the optimal column width in cortical organisation – too narrow and columns cannot develop distinct identities; too wide and columns fragment into subpatches.

5 Open Questions for Paper 33

- **Q22.** What is the attractor depth? Measure the recovery time after a deliberate zone-destroying perturbation (e.g., set $\mathbf{F} = 0$ for one zone at $T = 5000$). Faster recovery $\rightarrow$ deeper well.
- **Q23.** How does the optimal ZONE_W shift with κ and ν? Test: ZONE_W grid at $\kappa = 0.005$ and $\kappa = 0.080$. Prediction: optimal ZONE_W $\propto 1/\sqrt{\kappa}$ (diffusion sets coherence range).
- **Q24.** What is the copy-forward identity coherence length ℓ_{cf}? It is not the same as the spatial gradient length (Paper 31). Measure via zone identity persistence after zone boundary shifts.

6 Conclusion

Paper 32 establishes three results. First, the VCML zone attractor is a diffuse noisy attractor (CV=0.44), not a fixed point: zone identity is maintained stochastically via ongoing copy-forward reconstruction, and frequent transient collapses (zone swaps) are normal. Second, K_{eff} is a non-monotone function of ZONE_W with an optimum at $\approx$ 5–7 columns; the power-law description from Paper 30 was an artefact of limited sampling. Third, the $\nu = 0.003$ failure mode is fully recoverable with FA$\approx$ 2.0 ($\approx 10\times$ standard), confirming that the failure is energetic (insufficient consolidation) not structural (broken topology).

Taken together with Papers 28–31, these results confirm that the VCML zone attractor is a metastable stochastic state maintained by active reconstruction – qualitatively distinct from the stable fixed points produced by gradient-based learning.

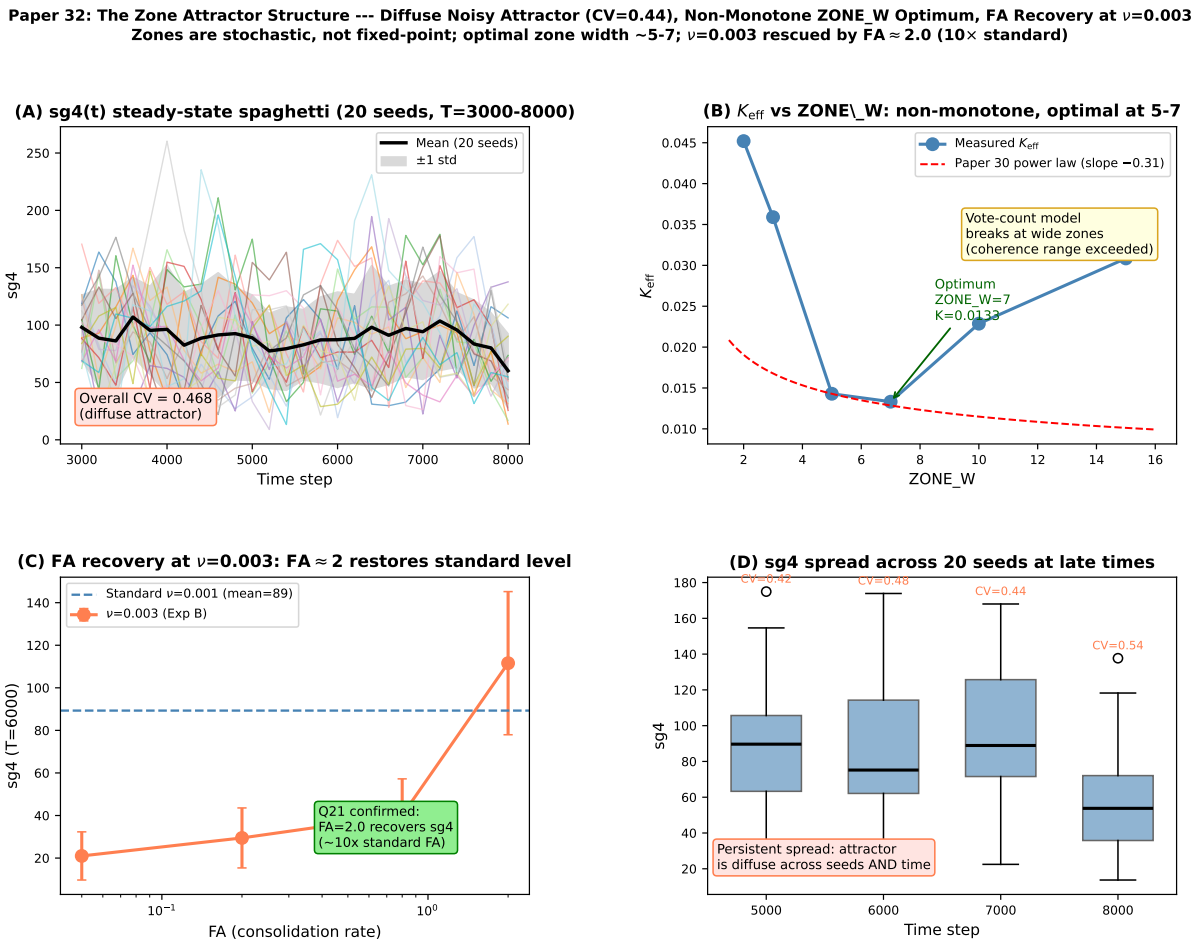

Figure 1: **Paper 32 results.** **(A)** sg4(t) spaghetti plot for 20 seeds at $T = 3000$–8000. High temporal variance (CV=0.44) and frequent transient drops indicate a diffuse noisy attractor. Mean $\pm$ 1 std shown in black. **(B)** K_{eff} vs ZONE_W: non-monotone curve with minimum at ZONE_W$\approx$ 5–7. The Paper 30 power-law prediction (dashed red) fits only the ascending flank. **(C)** sg4($T = 6000$) vs FA at $\nu = 0.003$: monotone recovery, reaching standard levels at FA$=$ 2.0. FA$=$ 5.0 caused numerical overflow. **(D)** Box plots of sg4 distribution across 20 seeds at 4 late time slices. Persistent spread (CV annotated) confirms that the attractor is diffuse at all times.